

# Opta Physical I/O Map

## Duplex Lift Station (DUPLEXLS)

ST program: logic/36\_duplex\_lift\_station.st  
MooreVIEW Opta firmware: 2.3.71  
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## Hardware

- Arduino Opta (base I1–I8 digital, I1–I6 analog, R1–R4 relay)
- Sequent D1608E digital expansion — slot 1 (X1\_I1...X1\_I16, X1\_R1...X1\_R8)
- Sequent A0602 analog expansion — slot 2 optional (X2\_AI1...X2\_AI8, X2\_PWM1...X2\_PWM4)

## Wiring notes

- Expansion DIs (+24 V active): input ON when float wet, run aux made, or fault contact active. Unwired inputs read OFF.
- Run feedback and fault inputs: +24 V active (input ON when aux made or fault contact active).
- Analog I1–I6: 0–1 V CT transmitters, linear 0–50 A (1.00 V = 50 A). Engineering tag = raw × 0.48828125.
- Relay outputs R1/R2 drive pump contactors from ST tags MOTOR1\_RUN / MOTOR2\_RUN.
- Float DIs: HIGH (X1\_I1) · LEAD (X1\_I2) · LAG (X1\_I3) · OFF (X1\_I4).
- PHASE\_FAULT (X1\_I7) clears auto pump demand in ST — same effect as dry off float while upper floats are on.

## Opta base — relay outputs

| Terminal / tag | Point       | Type | Driver channel | Logic tag  | Description      |
|----------------|-------------|------|----------------|------------|------------------|
| R1             | Relay out 1 | BOOL | R1             | MOTOR1_RUN | Pump 1 contactor |
| R2             | Relay out 2 | BOOL | R2             | MOTOR2_RUN | Pump 2 contactor |
| R3             | Relay out 3 | BOOL | R3             | —          | Spare            |
| R4             | Relay out 4 | BOOL | R4             | —          | Spare            |

## Opta base — digital inputs (spare)

| Terminal / tag | Point        | Type | Driver channel | Logic tag | Description                            |
|----------------|--------------|------|----------------|-----------|----------------------------------------|
| I7             | Digital in 7 | BOOL | I7             | —         | Spare (I1–I6 used as analog CT inputs) |
| I8             | Digital in 8 | BOOL | I8             | —         | Spare                                  |

## Opta base — analog inputs (CT 0–50 A)

| Terminal / tag | Point        | Type | Driver channel | Logic tag | Description                         |
|----------------|--------------|------|----------------|-----------|-------------------------------------|
| AI1 / I1       | Analog in 1  | REAL | I1_RAW         | AI1       | Motor 1 phase A amps                |
| AI2 / I2       | Analog in 2  | REAL | I2_RAW         | AI2       | Motor 1 phase B amps                |
| AI3 / I3       | Analog in 3  | REAL | I3_RAW         | AI3       | Motor 1 phase C amps                |
| AI4 / I4       | Analog in 4  | REAL | I4_RAW         | AI4       | Motor 2 phase A amps                |
| AI5 / I5       | Analog in 5  | REAL | I5_RAW         | AI5       | Motor 2 phase B amps                |
| AI6 / I6       | Analog in 6  | REAL | I6_RAW         | AI6       | Motor 2 phase C amps                |
| I7_RAW         | Analog raw 7 | INT  | I7_RAW         | —         | Spare raw (if I7 configured analog) |
| I8_RAW         | Analog raw 8 | INT  | I8_RAW         | —         | Spare raw (if I8 configured analog) |

## Expansion slot 1 — Sequent D1608E digital inputs

| Terminal / tag | Point | Type | Driver channel | Logic tag      | Description                       |
|----------------|-------|------|----------------|----------------|-----------------------------------|
| X1_I1          | DI 1  | BOOL | X1_I1          | LVL_HIGH       | High level float                  |
| X1_I2          | DI 2  | BOOL | X1_I2          | LVL_LEAD       | Lead float                        |
| X1_I3          | DI 3  | BOOL | X1_I3          | LVL_LAG        | Lag float                         |
| X1_I4          | DI 4  | BOOL | X1_I4          | LVL_OFF        | Off float                         |
| X1_I5          | DI 5  | BOOL | X1_I5          | P1_RUN_FB      | Pump 1 run feedback (fail-to-run) |
| X1_I6          | DI 6  | BOOL | X1_I6          | P2_RUN_FB      | Pump 2 run feedback (fail-to-run) |
| X1_I7          | DI 7  | BOOL | X1_I7          | PHASE_FAULT    | Phase fault                       |
| X1_I8          | DI 8  | BOOL | X1_I8          | GEN_RUN        | Generator running                 |
| X1_I9          | DI 9  | BOOL | X1_I9          | GEN_FUEL_FAULT | Generator fuel fault              |
| X1_I10         | DI 10 | BOOL | X1_I10         | GEN_FAULT      | Generator fault                   |
| X1_I11         | DI 11 | BOOL | X1_I11         | —              | Spare                             |
| X1_I12         | DI 12 | BOOL | X1_I12         | —              | Spare                             |
| X1_I13         | DI 13 | BOOL | X1_I13         | —              | Spare                             |
| X1_I14         | DI 14 | BOOL | X1_I14         | —              | Spare                             |
| X1_I15         | DI 15 | BOOL | X1_I15         | —              | Spare                             |
| X1_I16         | DI 16 | BOOL | X1_I16         | —              | Spare                             |

## Expansion slot 1 — Sequent D1608E relay outputs (spare)

| Terminal / tag | Point       | Type | Driver channel | Logic tag | Description                           |
|----------------|-------------|------|----------------|-----------|---------------------------------------|
| X1_R1          | Relay out 1 | BOOL | X1_R1          | —         | Spare (pump contactors on base R1/R2) |
| X1_R2          | Relay out 2 | BOOL | X1_R2          | —         | Spare                                 |
| X1_R3          | Relay out 3 | BOOL | X1_R3          | —         | Spare                                 |
| X1_R4          | Relay out 4 | BOOL | X1_R4          | —         | Spare                                 |
| X1_R5          | Relay out 5 | BOOL | X1_R5          | —         | Spare                                 |
| X1_R6          | Relay out 6 | BOOL | X1_R6          | —         | Spare                                 |
| X1_R7          | Relay out 7 | BOOL | X1_R7          | —         | Spare                                 |
| X1_R8          | Relay out 8 | BOOL | X1_R8          | —         | Spare                                 |

## Expansion slot 2 — Sequent A0602 analog (optional RTD)

| Terminal / tag | Point    | Type | Driver channel | Logic tag   | Description                        |
|----------------|----------|------|----------------|-------------|------------------------------------|
| X2_AI1         | Analog 1 | REAL | X2_AI1         | MOTOR1_TEMP | Motor 1 winding temp (cal in Tags) |
| X2_AI2         | Analog 2 | REAL | X2_AI2         | MOTOR2_TEMP | Motor 2 winding temp (cal in Tags) |

|        |          |      |        |   |       |
|--------|----------|------|--------|---|-------|
| X2_AI3 | Analog 3 | REAL | X2_AI3 | — | Spare |
| X2_AI4 | Analog 4 | REAL | X2_AI4 | — | Spare |
| X2_AI5 | Analog 5 | REAL | X2_AI5 | — | Spare |
| X2_AI6 | Analog 6 | REAL | X2_AI6 | — | Spare |
| X2_AI7 | Analog 7 | REAL | X2_AI7 | — | Spare |
| X2_AI8 | Analog 8 | REAL | X2_AI8 | — | Spare |

## Expansion slot 2 — Sequent A0602 PWM outputs (spare)

| Terminal / tag | Point | Type | Driver channel | Logic tag | Description |
|----------------|-------|------|----------------|-----------|-------------|
| X2_PWM1        | PWM 1 | REAL | X2_PWM1        | —         | Spare       |
| X2_PWM2        | PWM 2 | REAL | X2_PWM2        | —         | Spare       |
| X2_PWM3        | PWM 3 | REAL | X2_PWM3        | —         | Spare       |
| X2_PWM4        | PWM 4 | REAL | X2_PWM4        | —         | Spare       |

## Related documents

Live values: MooreVIEW Tags !' Live I/O, or Opta <http://<opta-ip>/io-map> when on Ethernet.

ST logic reference: [st/logic/36\\_duplex\\_lift\\_station.st](st/logic/36_duplex_lift_station.st) (PHYSICAL I/O MAP header).